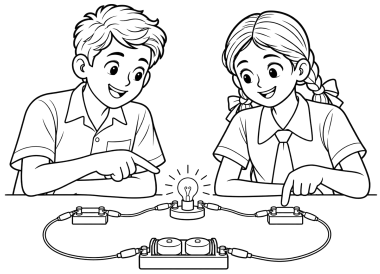
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 4
Use of Microcontrollers



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 4 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Activities 4.1 - 4.6, safety, Arduino programs, a quiz
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 4 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links and notes, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ micro:bits, micro USB cables and the block editor are ready on the computers.
- ☐ An Arduino Uno, a breadboard, LEDs, wires and the two sensors are ready.
- ☐ The Arduino software is installed for Practicals 4 - 6.

"Students who build it with their own hands never forget how it works."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Match the components	6
	B Fill in the blanks	8
	C Inputs and outputs of devices	6
	D True or false	8
	E Input, process or output?	8
	F SBC or SCC?	4
	G Choose the correct answer	6
Chapter 2	A Fill in the blanks	8
	B Which part do you use?	8
	C Label the micro:bit	11
	D True or false	6
Chapter 3	A Draw the LED patterns	8
	B Label the block editor	7
	C Basic or Input?	8
	D Answer these	4
	E What will the micro:bit show?	8
	F Match the blocks	6
Chapter 4	A Label the Arduino board	7
	B Label the sensors	9
	C Are they connected?	6
	D PIR or ultrasonic?	8
	E Crossword	10
End	Unit 4 Test	40
TOTAL		**200**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Start the unit outside: look at a street lamp near the school and ask who switches it on and off.
- Show the Sinhala editor (More, Language, Sinhala) to students who find English hard.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (46)	Ch 2 (33)	Ch 3 (41)	Ch 4 (40)	Test (40)	Total (200)
1							
2							
3							
4							
5							
6							
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27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

Workbook Activities 4.1 - 4.6 of the textbook

Practical	Students do	Check that ...
1 (Act. 4.1)	find devices with sensors; fill the input / output table	the input is a sensor or the buttons, the output is what the device does, and the microcontroller is the process.
2 (Act. 4.2)	a first program with the Basic blocks; Download	the name scrolls once (on start); the heart and the 9 repeat (forever); the file is saved on the MICRO BIT drive.
3 (Act. 4.3)	Input blocks: buttons A and B, shake, light level	A shows the heart, B shows Bye!, a shake shows the temperature; a hand over the board makes the light level smaller.
4 (Act. 4.4)	an LED on a breadboard with the Arduino Uno	the longer leg (+) goes to the digital pin (with a resistor), the shorter leg (-) to GND; the LED blinks.
5 (Act. 4.5)	a PIR sensor switches the LED	VCC to 5V, GND to GND, OUT to a digital pin; the LED lights when someone moves in front of the sensor.
6 (Act. 4.6)	an ultrasonic sensor measures a distance	VCC to 5V, GND to GND, TRIGGER and ECHO to two digital pins; the distance falls as the book comes nearer.

! SAFETY AND MATERIALS

- micro:bit: one board and one micro USB cable per pair, computers with the block editor (makecode.microbit.org). A 3 V battery pack is optional.
- Arduino: an Arduino Uno with its USB cable, a breadboard, LEDs with a 220 ohm resistor, jumper wires, a PIR sensor module and an ultrasonic sensor module (often sold as HC-SR501 and HC-SR04).
- Only you connect the boards to the computer. Use the USB cable (5 V) - never mains electricity.

> PROGRAMS FOR PRACTICALS 4 - 6 (ARDUINO SOFTWARE)

The textbook does not print the Arduino programs. Type these short ones in the Arduino software and upload them; the distance of Practical 6 appears in the Serial Monitor.

Practical 4: the LED on pin 8

```
void setup() {
  pinMode(8, OUTPUT);
}
void loop() {
  digitalWrite(8, HIGH);
  delay(1000);
  digitalWrite(8, LOW);
  delay(1000);
}
```

Practical 5: PIR OUT on pin 2

```
void setup() {
  pinMode(2, INPUT);
  pinMode(8, OUTPUT);
}
void loop() {
  digitalWrite(8, digitalRead(2));
}
```

Practical 6: TRIGGER 9, ECHO 10

```
void setup() {
  pinMode(9, OUTPUT);
  pinMode(10, INPUT);
  Serial.begin(9600);
}
void loop() {
  digitalWrite(9, HIGH);
  delayMicroseconds(10);
  digitalWrite(9, LOW);
  long t = pulseIn(10, HIGH);
  Serial.println(t * 0.034 / 2);
  delay(500);
}
```

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Sensors and microcontrollers	47 - 50, 65	4.1 (p. 49)
Chapter 2 - Microcontroller kits and the micro:bit	51 - 53	-
Chapter 3 - Coding the micro:bit with blocks	54 - 58	4.2, 4.3 (p. 59)
Chapter 4 - Arduino, breadboards and sensors	59 - 64	4.4 - 4.6 (p. 64)
Unit 4 in One Page	65 (summary)	-

(?) QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What are the basic functions of a computer?	**Input, process, output**
2. What is a microcontroller?	**A chip that works by instructions**
3. Name the four components of a microcontroller.	**CPU, memory, clock, pins**
4. Which memory stores the program?	**Non-volatile memory**
5. Who developed the micro:bit?	**The BBC**
6. How many LED bulbs does the micro:bit have?	**25**
7. Which block runs again and again?	**forever**
8. Which Arduino board do we use?	**Arduino Uno**
9. Why do we use a breadboard?	**No welding is needed**
10. Which sensor works like a bat?	**The ultrasonic sensor**

(>) PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Use the micro:bit block editor of Practicals 2 and 3. The name scrolls once; A and B work; the light level changes when the board is covered.

Step	The student must ...	Marks
1	Open the block editor and start a new project.	2
2	In **on start**, show your name with **show string**.	3
3	**on button A pressed**: show a heart. **on button B pressed**: show the number 5.	6
4	In **forever**, show the **light level**, then **pause (ms)** 1000.	5
5	Test the program on the micro:bit model, then download it to the micro:bit.	3
6	Save the project with your name.	1
	Total	**20**

(★) TEACHING TIPS

- Write INPUT, PROCESS and OUTPUT on the board and keep them there for the whole unit.
- Bring a broken toy, remote controller or calculator and show the microcontroller chip inside it.

ANSWERS

Chapter 1

Sensors and microcontrollers

A Match the components

6 marks

1. CPU → E	**2.** Non-volatile memory → F	**3.** Volatile memory → D
4. System clock → A	**5.** Peripheral devices → C	**6.** Microcontroller → B

B Fill in the blanks

8 marks

1. process	**2.** sensor	**3.** chip	**4.** non-volatile
5. RAM	**6.** clock	**7.** digital	**8.** dark

C Inputs and outputs of devices

6 marks

Device	Input	Output
Solar lamp	The light sensor detects darkness (or light).	The lamp turns on (or off) automatically.
Washing machine	The user presses buttons to give instructions.	The machine washes the clothes.
Microwave oven	The user gives the temperature and the duration.	It holds the heat for that period and stops.

1 mark for each input and each output. Accept other sensible words.

D True or false

8 marks

1. **T - Sensors can measure soil conditions and temperature.**
2. **F - A microcontroller can work without any program instructions.**
3. **F - Data in non-volatile memory is erased when there is no electricity.**
4. **T - The data and instructions for the CPU are held in the volatile memory.**
5. **T - A microcontroller has four basic components.**
6. **F - The peripheral pins collect data only in digital form.**
7. **T - A washing machine works according to the buttons the user presses.**
8. **F - A Raspberry pi is a single chip computer.**

TEACHING TIPS

- Say it often: the sensor is the input, the microcontroller is the process, the bulb is the output.
- Ask students to say what a program will do before they press Download. Then test it.

E Input, process or output?

8 marks

Which part is it?	Input	Process	Output
The light sensor detects that it is getting dark.	☑	☐	☐
The microcontroller decides to switch on the street lamp.	☐	☑	☐
The street lamp lights up.	☐	☐	☑
The user presses the buttons of the washing machine.	☑	☐	☐
The microcontroller follows the program instructions.	☐	☑	☐
The microwave oven stops heating at the set time.	☐	☐	☑
The light sensor of a solar lamp detects sunlight.	☑	☐	☐
The CPU processes the data into information.	☐	☑	☐

1 mark each. The sensor and the buttons are inputs; the microcontroller and its CPU do the process; the lamp, the oven and the bulb are outputs.

F SBC or SCC?

4 marks

1. All the parts are on one circuit board. - **SBC**	**2.** The CPU, input, output and memory are in one IC. - **SCC**
3. A Raspberry pi - **SBC**	**4.** The Arduino chip - **SCC**

G Choose the correct answer

6 marks

1. (c) four	**2.** (c) system clock
3. (a) non-volatile memory	**4.** (c) solar lamp
5. (a) duration	**6.** (d) the bulb

★ TEACHING TIP

- Teach the two memories with a picture: non-volatile is a notebook, volatile (RAM) is a whiteboard.

ANSWERS

Chapter 2

Microcontroller kits and the micro:bit

A Fill in the blanks

8 marks

1. BBC	**2.** 25	**3.** USB	**4.** storage
5. 3	**6.** embedded	**7.** B	**8.** kits

B Which part do you use?

8 marks

1. To connect the micro:bit to a computer → **Micro USB port**	**2.** To give an input by pressing it → **Buttons A and B**
3. To show a pattern, a number or a word → **LED bulbs**	**4.** To connect a 3 V external power supply (batteries) → **Battery connector**
5. To connect devices using Bluetooth and send radio waves → **Bluetooth antenna**	**6.** To find the direction and the motion → **Compass and accelerometer**
7. To process the data → **CPU**	**8.** For digital and analog inputs and outputs → **Pins 0, 1 and 2**

C Label the micro:bit

11 marks

1. Buttons A and B	**2. Pins 0, 1 and 2 (input / output pins)**	**3. LED bulbs**
4. Port to connect power (3V)	**5. Ground port (GND)**	**6. Bluetooth smart antenna**
7. CPU (Central Processing Unit)	**8. Micro USB port**	**9. Battery connector (3 V)**
10. Accelerometer and compass	**11. Pin edge connector**	

1 mark each. 1 - 5 are on the front (Figure 4.1), 6 - 11 on the back (Figure 4.2). Accept "3V" for 4, "GND" or "ground back port" for 5 and "antenna" for 6.

D True or false

6 marks

1. **T - The micro:bit has the basic features of a computer.**
2. **F - Extra LED bulbs must be added to the micro:bit to show a pattern.**
3. **T - The micro:bit can connect to other devices using Bluetooth.**
4. **F - The CPU is on the front of the micro:bit.**
5. **T - The micro:bit module is about 4 cm high.**
6. **F - In Figure 4.4 the micro:bit is shown as Local Disk (C:).**

TEACHING TIPS

- Check every micro:bit, cable and battery pack before the lesson. A bad cable wastes a whole period.
- Not enough micro:bits? The micro:bit model on the screen runs every program of Practicals 2 and 3.
- Let pairs share one board: one student drags the blocks, the other one checks and downloads.
- Remind students: work that must repeat goes into forever, work that happens once goes into on start.

ANSWERS

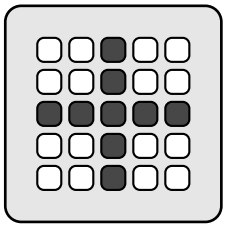
Chapter 3

Coding the micro:bit with blocks

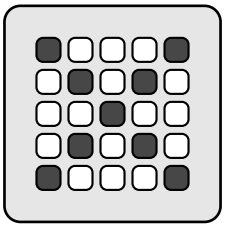
A Draw the LED patterns

8 marks

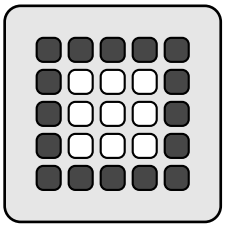
a plus sign



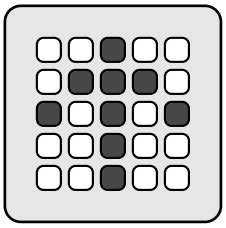
a big X



a square on the edge



an arrow (North)



2 marks each. Accept any clear pattern of the right shape (for example a smaller square or a plus sign made of 5 LEDs).

B Label the block editor

7 marks

1. micro:bit model	**2. Projects**	**3. Blocks / JavaScript**	**4. Block editor menu**
5. on start and forever blocks	**6. Download**	**7. Name of the project (Untitled)**	

1 mark each (Figure 4.7). Accept "the micro:bit board on the screen" for 1 and "the menu" for 4.

C Basic or Input?

8 marks

The block	Basic	Input
show number	☑	☐
on button A pressed	☐	☑
show leds	☑	☐
light level	☐	☑
pause (ms)	☑	☐
on shake	☐	☑
clear screen	☑	☐
temperature (°C)	☐	☑

TEACHING TIP

- Use the LED grids: let students design a pattern on paper first, then build it with show leds.

D Answer these

4 marks

1. Why can even a beginner code the micro:bit? (2 marks)

The block editor has a drag-and-drop facility: we join blocks instead of typing a program (1 mark). It can even be used in Sinhala (1 mark).

2. Name two other ways (languages) to code the micro:bit. (2 marks)

Any two of JavaScript, Python and C++ (1 mark each).

E What will the micro:bit show?

8 marks

1. **The number 7 is shown once, when the micro:bit starts.**
2. **The word Hi! is shown when button A is pressed.**
3. **A heart is shown when the micro:bit is shaken.**
4. **The temperature is shown again and again (all the time).**

2 marks each: 1 mark for what is shown, 1 mark for when it is shown (once, when a button is pressed, when it is shaken, again and again).

F Match the blocks

6 marks

1. show string → C	**2.** show leds → E	**3.** pause (ms) → F
4. clear screen → B	**5.** on pin P0 pressed → D	**6.** compass heading (°) → A

TEACHING TIPS

- Only the teacher connects the Arduino to the computer in Practicals 4 - 6. Never use mains power.
- Put the LED in the right way round: the longer leg is the + pin. A wrong LED simply stays off.

ANSWERS

Chapter 4

Arduino, breadboards and sensors

A Label the Arduino board

7 marks

1. Mini USB	**2. Power regulator**	**3. DC power**	**4. Power pins**
5. Analog pins	**6. ATMEGA328p microcontroller**	**7. Digital pins**	

1 mark each (Figure 4.12, Table 4.1). Accept "electric controller" for 2 and "DC power supply" for 3.

B Label the sensors

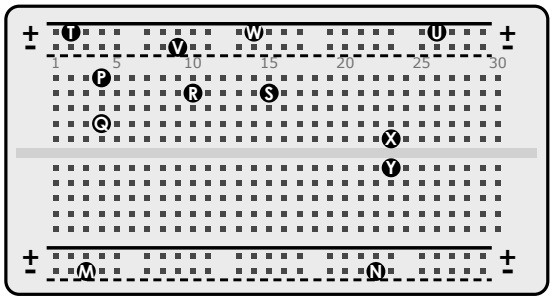
9 marks

1. GND	**2. OUT**	**3. VCC**
4. Delay adjust	**5. Sensitivity adjust**	**6. VCC**
7. TRIGGER	**8. ECHO**	**9. GND**

1 mark each. PIR (Figure 4.15): pins GND, OUT, VCC and the two trimpots. Ultrasonic (Figure 4.18): VCC, TRIGGER, ECHO, GND.

C Are they connected?

6 marks



1. **P and Q: Yes (the same column of five)**
2. **R and S: No (different columns)**
3. **T and U: Yes (the same + row)**
4. **V and W: No (a + row and a - row)**
5. **X and Y: No (across the middle groove)**
6. **M and N: Yes (the same - row)**

D PIR or ultrasonic?

8 marks

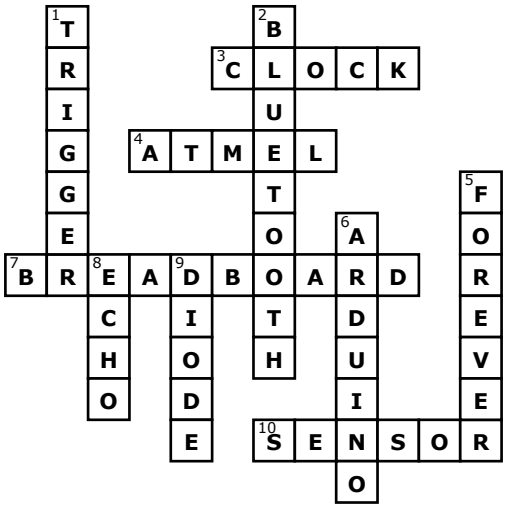
The sensor ...	PIR	Ultrasonic
It estimates the distance from it to an object.	☐	☑
It measures infrared objects, like humans, in its field of view.	☑	☐
It has a TRIGGER pin and an ECHO pin.	☐	☑
Its OUT pin outputs 3.3 V.	☑	☐
It works like a bat finding its route.	☐	☑
Its sensitivity can be adjusted up to 7 m.	☑	☐
It sends ultrasound signals and reads the reflected signals.	☐	☑
It has a trimpot to adjust the time delay of the output.	☑	☐

★ TEACHING TIPS

- Use a 220 or 330 ohm resistor in series with the LED, so the LED and the pin are safe.
- For the PIR sensor, wait about a minute after power on - it needs time to settle.
- Show the bat video of any nature programme before the ultrasonic sensor. Students remember echoes.
- Keep the parts in labelled boxes: boards, sensors, LEDs, wires. Count them at the end of the lesson.
- Let students swap books and check the labels of a friend before you mark them.
- Give a five-question oral quiz at the start of every lesson. Use the list on the practicals page.

E Crossword

10 marks



Across: 3 CLOCK, 4 ATMEL, 7 BREADBOARD, 10 SENSOR

Down: 1 TRIGGER, 2 BLUETOOTH, 5 FOREVER, 6 ARDUINO, 8 ECHO, 9 DIODE

MARKING SCHEME

Unit 4 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (a) sensors	**2.** (c) non-volatile memory
3. (b) Raspberry pi	**4.** (c) 25
5. (b) forever	**6.** (a) 1 second
7. (a) ATMEGA328p	**8.** (b) ultrasonic

2 Fill in the blanks

8 marks (1 each)

1. output	**2.** synchronizes	**3.** RAM	**4.** USB
5. on start	**6.** breadboard	**7.** one	**8.** 3.3

3 Name the pictures

8 marks (1 each)

1. micro:bit	**2. Arduino Uno**	**3. breadboard**	**4. LED**
5. PIR sensor	**6. ultrasonic sensor**	**7. solar lamp**	**8. washing machine**

Raspberry pi and microwave oven are not used.

TIP: Students who finish early can colour the pictures, or design a new LED pattern for a friend.

4 Answer these

16 marks

1. Name the four basic components of a microcontroller. (4 marks)

CPU, memory, system clock and peripheral devices. 1 mark each.

2. A solar lamp turns on when it gets dark. Write its input, process and output. (3 marks)

Input: the light sensor detects darkness. Process: the microcontroller processes the data. Output: the lamp turns on. 1 mark each.

3. Which block would you use: (a) to display a number (b) to erase the displayed data (c) to show a heart? (3 marks)

(a) show number (b) clear screen (c) show icon. 1 mark each.

4. What is the difference between the on start block and the forever block? (2 marks)

on start holds the initial action (it runs once, at the start) (1 mark); forever holds the repetitive work (it runs again and again) (1 mark).

5. Write the four pins of the ultrasonic sensor. Which pin gives the output? (2 marks)

VCC, TRIGGER, ECHO and GND (1 mark). ECHO gives the output (1 mark).

6. What is a single chip computer? Give one example. (2 marks)

The CPU, input, output and memory are all built into a single integrated circuit (IC) (1 mark). Example: the Arduino chip (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They call the microcontroller the input. The **sensor** is the input; the microcontroller does the process.
- They mix up the memories: **non-volatile** keeps the program without electricity; **volatile** (RAM) loses its data.
- They put repeated work into on start. Work that must repeat goes into **forever**.
- They use show string for numbers, or show number for words. Numbers: show number; words: show string.
- They say the CPU and the micro USB port are on the front. Both are on the **back** of the micro:bit.
- They think holes across the middle groove of a breadboard are connected. They are not.
- They join the LED the wrong way. The **longer leg** is the + pin.
- They mix up TRIGGER and ECHO. TRIGGER is for **input**; ECHO gives the **output**.

TEACHING TIP

- Link the unit to farming: sensors for soil and temperature can help Sri Lankan farmers.

NOTES ON THE TEXTBOOK (PAGES 47 - 65)

- p. 50 "Arduino clip" means the Arduino **chip**; p. 51 "Arduano" means Arduino.
- p. 51 the websites are www.arduino.cc and www.raspberrypi.org (printed "ardino" and "raspberry").
- p. 55 the editor is at **makecode.microbit.org** (printed "www.makecode.micro:bit.org").
- p. 57 show arrow displays an **arrow** (printed "to display an array").
- Part 2 of the Arduino is the "Power regulator" in Figure 4.12 and the "Electric controller" in Table 4.1.
- p. 62 "welding" means joining with hot metal (soldering). Breadboards need neither.
- The textbook calls the Arduino port "Mini USB"; the Uno has a full-size USB (type B) port - accept both.

HOMEWORK AND EXTRA PRACTICE

- List five devices at home that use a sensor or buttons and a microcontroller. Write their input and output.
- Draw the front and the back of the micro:bit from memory and label the 11 parts.
- Plan a micro:bit program on paper: button A shows a smiley (show leds), button B clears the screen.
- Fast learners: try the Music or the Led menu in the block editor and show the class what they do.
- Draw the Arduino Uno board and label its seven parts (Figure 4.12, Table 4.1).

TEACHING TIPS

- Praise careful wiring. A neat breadboard is easy to check and easy to correct.
- Mark the unit test together the next day. Students learn most from their own mistakes.
- Ask: what is the input, what is the process, what is the output? for every device you talk about.
- Let students hold a real chip. Count its pins together and find the small dot or notch.
- Show the micro:bit drive in the Computer window. Students like to see the board as a storage unit.
- Before Practical 2, show how to drag a block into on start and how to delete a block.
- If a program does not work, read the blocks aloud from top to bottom. Most mistakes show up at once.

TIP: Shake, press A, press B: let students guess what each program does before they test it.